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Keynesian model building and the rational expectations critique

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1. Introduction

Until the early 1970's, the economists who opposed the Keynesians had to be content with pulling a few fish off their opponents hooks. But when what has become known as the theory of 'rational expectations' began to be developed, these economists found that they could simply dynamite all the fish in the lake... Today ... a fleet of stunned Keynesians is quibbling about which of their few remaining fish are still flopping (Willes, 1980, p. 81).

In recent years Keynesian analysis and economic modelling have come in for increased criticism, particularly from those economists who advocate the rational expectations hypothesis—the so-called new classical school. The objective of this paper is to assess the more influential of these criticisms.

The paper is divided into separate but related parts. Section 2 examines arguments about expectation formation, whilst Section 3 concentrates on broader arguments about the methodological underpinnings of economic modelling. The two parts are related because theories concerning the formation of expectations underpin the new classical economists' criticisms of Keynesian modelling methodology.

The principle thesis put forward in both parts of the paper is that economists in the new classical school habitually criticise views and theories which are not really Keynesian at all. It is argued, moreover, that many of the points made by the new classical economists are well understood by Keynesians and are in fact a part of wider ranging criticisms that Keynesians make of other contemporary modelling practices.

The paper does not attempt a detailed justification of Keynesian positions on the issues considered. Rather it outlines Keynesian views on these issues in order to justify the contention that they are generally very different from positions typically but erroneously attributed to Keynesians by advocates of the rational expectations hypothesis. Nor does the paper attempt a general critique of the rational expectations approach to modelling. It considers only some common criticisms that advocates of the rational expectations hypothesis level at Keynesian analysis and modelling practice. However, given that alleged weaknesses in Keynesian modelling are widely supposed to be a valid

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justification for adopting the rational expectations approach, this focus is not unimportant.[1]

2. Theories and models of expectations

According to the rational expectations hypothesis people form expectations as if they know (and utilise their knowledge of) the process which will ultimately generate the actual outcomes (see Muth, 1961, for the defining article and see, for example, B. Friedman, 1979, and Shiller, 1978, for useful discussions). The outstanding feature of this hypothesis is not that people are assumed to make, in some sense 'optimal' use of the information available to them (this is hardly contentious or new—see below), but rather the amount and nature of information that people supposedly hold and use in the process of forming expectations.

Attempts to justify the hypothesis are frequently made by contrasting it with an alternative hypothesis or model which makes limited information assumptions, and is highly simplistic. This alternative model is the following, supposedly Keynesian, adaptive expectations hypothesis:

$$ {}_{t}Y^{e}_{t+1} = {}_{t-1}Y^{e}_{t} + \lambda(Y_t - {}_{t-1}Y^{e}_{t}) \quad \lambda \text{ constant}, 0<\lambda<1 \tag{2.1} $$

where ${}_{t}Y^{e}_{t+1}$ represents the expectation of the variable Y for time $t+1$ with the expectation formed and held at time t (see for example Willes, 1980).

The model possesses two frequently highlighted limitations. The first is the property that when the variable Y reveals a trend, the expectations generated by the model are certain to be consistently inaccurate. This can most easily be seen by rewriting the model in the form:

$$ {}_{t}Y^{e}_{t+1} - {}_{t-1}Y^{e}_{t} = \lambda(Y_t - {}_{t-1}Y^{e}_{t}) \quad 0<\lambda<1 \tag{2.2} $$

If Y_t reveals a trend, then for expectations also to do so the left-hand side of (2.2) must be different from zero, in which case the right-hand side must also be different from zero, thus preventing ${}_{t-1}Y^{e}_{t}$ from being equal to Y_t. (This criticism of the model is made, for example, in Barro and Fischer, 1976, p. 155; Minford, 1980, p. 10 and M. Friedman, 1975, p. 27.)

A second limitation concerns the assumed constancy of the adjustment coefficient λ. Many economists have suggested that λ will be closer to unity at times of greater uncertainty when people are assumed to attach greater weight to actual observations when deriving forecasts. The new classical economists argue that the inflexibility of (2.1) prevents the consistent or 'valid' econometric testing of the core monetarist assumption, the natural rate hypothesis. (The earliest development of this argument is in Lucas, 1972, but see also M. Friedman, 1975, p. 26; Minford, 1980, p. 10 or Sargent and Wallace, 1975.)

[1] Although the new classical economists typically refer to Keynesian analysis and modelling as though the Keynesian position was unique and well-defined, there obviously exists some variation in the views and practices that may be considered to be Keynesian in their approach. The Keynesian interpretations adopted here are inevitably those most familiar to the author within what may loosely be described as the Cambridge (UK) tradition. They may not be completely endorsed, therefore, by all who consider their own positions to be Keynesian—particularly those strongly influenced by the 'neo-classical synthesis'. (Minford has argued that the main 'forecasting' models outside Cambridge are, in fact, neo-Keynesian—by which is meant that they 'have been modified in the light of a variety of research findings inspired largely by "monetarism" . . .' (Minford, 1980, p. 4).) Certainly the interpretations presented here represent the views and positions maintained by the author.

There also exists a third criticism levelled at model (2.1) which for expositional purposes we shall refer to as an objection to (as opposed to a limitation of) the hypothesis. This is that the model assumes that only information on past values of Y conditions expectations of current and future values. Advocates of the rational expectations hypothesis typically argue that model (2.1) implicitly ignores the costs or ease of obtaining other relevant information (see for example Willes, 1980, p. 286).

The objective of this section is to examine the relevance of these criticisms for Keynesian analysis and modelling practice. Given that model (2.1) is sometimes attributed to Keynes directly we first assess the legitimacy of making such an association. Then we briefly explore the relation of the simple adaptive hypothesis to the assumptions of Keynesian forecasting models and we also discuss the criticism that Keynesian expectations assumptions, even if not the same as model (2.1), ignore the costs or ease of obtaining relevant information. Finally some implications of the discussion are drawn out.

Keynes' views on expectations

It is not the intention to include here a detailed account of Keynes' views on expectation formation (the interested reader should read Keynes, 1973A, pp. 147–164; Keynes, 1973B and Keynes, 1973C, pp. 336–339) but merely to examine the legitimacy of associating Keynes' views with model (2.1).[1]

In the *General Theory*, Keynes discusses expectations mainly, but not exclusively, in the context of the expected returns to 'entrepreneurial activity'. In this context Keynes distinguishes short-term expectations (where capital is fixed and expectations relate to returns for finished output using existing plant at the time when the commitment to produce is made) from long-term expectations (which are concerned with what the producer can hope to earn by adding to existing capital equipment).

Focusing on the latter, Keynes argues that the 'outstanding fact' relevant to the formation of long-term expectations is the 'extreme precariousness of the basis of knowledge' on which such expectations are formed, and he suggests that in many cases 'it amounts to little or sometimes to nothing' (p. 150). As a result Keynes believed that people in forming expectations probably fall back on the convention of assuming the existing state of affairs will continue indefinitely unless there is a specific reason for expecting a change (p. 148).

Of course, Keynes accepts that the adoption of such a convention has its 'weak points', and its precariousness is viewed as largely responsible for insufficient investment taking place. He notes the separation of ownership and control (and thus of power and understanding), the absurd influence of the everyday fluctuations of profits of existing investments on the market, and the sensitivity of opinion of 'ignorant individuals' to 'factors which do not really make much difference to the prospective yield' as among factors responsible for this precariousness (p. 153). Moreover in addition to instability resulting from speculation Keynes argues that there is also 'instability due to the characteristic of human nature that a large proportion of our positive activities depend on spontaneous optimism' (p. 161), or what he called 'animal spirits'.

However in highlighting these factors Keynes does not suggest that all investment activity depends on waves of irrational psychology:

[1]For interesting discussions which focus more upon the importance of Keynes' views on expectations in underpinning the analysis in the *General Theory* (and also the *Treatise on Money*) see, for example, Shackle, 1965, pp. 43–96; Shackle, 1967; Shackle, 1972; and Shackle, 1974.

On the contrary the state of long-term expectation is often steady... We are merely reminding ourselves that human decisions affecting the future, whether personal or political or economic, cannot depend on strict mathematical expectation, since the basis for making such calculation does not exist ... (Keynes, 1973A, p. 163).

Whilst this somewhat schematic outline of Keynes' view on long-term expectations clearly reveals an explicit attempt to repudiate (at least for certain domains) what has since become known as the rational expectations hypothesis, it is also clear that, whatever else emerges, the view put forward is not in any way consistent with the simple adaptive expectations model defined above. Indeed Keynes explicitly argues that 'it is the nature of long-term expectations that they cannot be checked at short intervals in the light of realised results' (p. 51).

Not surprisingly, therefore, attempts made to associate *specific* statements of Keynes with model (2.1) have focused on passage(s) which relate instead to Keynes' views on short-term expectation formation. The association is usually inferred from the following passages (e.g. Beenstock, 1980, p. 37):

In practice the process of revision of short-term expectation is a gradual and continuous one, carried out largely in the light of realised results; so that expected and realised results run into and overlap one another in their influence... The most recent results usually play a predominant part in determining what these expectations are (Keynes, 1973A, p. 50).

Moreover, the expectation of future consumption is so largely based on current experience of present consumption that a reduction in the latter is likely to depress the former (Keynes, 1973A, p. 210).

It is evident, however, that these passages do not necessarily link Keynes' view with model (2.1); the most that can be deduced is that Keynes believed short-term expectations to be adaptive in some way. In fact, in the first passage, the use of the word 'usually' suggests that he thought that recent results may not *always* play the predominant part in determining current expectations, and therefore the weight attached to recent results is not constant as model (2.1) suggests. More important however, the *General Theory* includes further passages on short-term expectations which, as we illustrate below, are consistent with the construction of a more general adaptive model which is different from model (2.1) in a manner which makes it immune to the limitations of the latter outlined above.

Expectations formation in a Keynesian macro-model

Although the adaptive hypothesis embodied in equation (2.1) is frequently associated with Keynesian econometric modelling it is rarely the case either that it is associated with any specific model, or indeed that the expectation assumptions of Keynesian macro-models are examined explicitly at all. A brief reference to the expectational assumptions in a specific Keynesian model may be helpful. For this purpose we concentrate on the Cambridge Multisectoral Model of the UK economy, because its expectational assumptions are reasonably explicit and also because it is the model with which the author is most familiar. (For a general description of the model see, for example, Barker *et al.*, 1980.)

For certain variables, expectations in this model *are* assumed to be formed adaptively. The hypothesis adopted is, however, different from that underlying equation (2.1),

due mainly to the fact that various forms of uncertainty are modelled explicitly.[1]

The outcome is an adaptive model which is reasonably general, and is flexible in the sense that the adjustment coefficient is time varying in an *a priori* acceptable fashion; thus making the model immune to the limitations of equation (2.1) discussed above. (The formal results are set out in the appendix to this paper. It will also be clear from this appendix that the model is completely consistent with the usual notion of 'optimality' with respect to the use of information.)

The intention here is not to labour the desirability or otherwise of this more general model. Indeed as with the rational expectations hypothesis it cannot even be demonstrated that this model is unambiguously a better approximation to 'reality' than equation (2.1). It does, however, serve to illustrate that the limitations of model (2.1) can be overcome by a variety of alternative hypotheses. The adoption of the rational expectations hypothesis is by no means necessary in order to overcome such limitations.

Relevant information

Whilst Keynesian views on expectations are not bound by the obvious limitations of model (2.1) they may appear more vulnerable to the third criticism—or 'objection'—that in the formation of expectations only the past history of a variable is utilised even though other relevant information may be available. Members of the new classical school, by contrast, argue for the assumption that the whole structure of economic relationships and the out-turns of all relevant forcing variables are always essentially known and used when expectations are formed. Clearly, disagreements concerning the nature of knowledge and how it is acquired are hardly new and no attempt to discuss the general issues can be made here.[2] But two relevant points should be noted.

Firstly it is misleading to dismiss either adaptive hypotheses or Keynes' expectational assumptions which rely heavily on the past history of the relevant variable on the grounds that they ignore the costs or ease of obtaining relevant information. Indeed for Keynes, it is precisely factors of this kind which underlie the adoption of particular kinds of expectational assumptions. We have seen above that Keynes' implicit rejection of the rational expectations hypothesis for longer-term expectations rests upon the 'outstanding fact' of the 'extreme precariousness' of the basis of knowledge on which such expectations are formed. His views concerning short-term expectations were based on similar considerations:

> It would be too complicated to work out the expectations *de novo* whenever a productive process was being started; and it would, moreover, be a waste of time since a large part of the circumstances usually continue substantially unchanged from one day to the next. Accordingly it is sensible to base expectations on the assumption that the most recently realised results will

[1]The uncertainty assumptions, moreover, are consistent with the views of Keynes, if not explicitly made by him. Thus he suggests that an entrepreneur 'does not entertain a single undoubting expectation of what the sale proceeds of a given output will be but several hypothetical expectations held with varying degrees of probability and definiteness . . .' (Keynes, 1973A, p. 24). Keynes explicitly accepts the assumption of subjective probabilities for short-term expectations although not necessarily for all outcomes. Even if probabilities are not attached to all outcomes, however, this may partly reflect the nature of the variable over which expectations are being formed. As we saw above, Keynes also discusses adaptive expectations in the context of disposable income or consumption. If this is estimated in real terms net of taxation and other, possibly 'transient', influences, the more general assumption seems appropriate.

The second type of uncertainty assumption required to generate the adaptive model is the uncertainty relating to the assumed relationship between current and future values of a variable. This is also explicit in Keynes; in fact one of the hypothesised 'principle objective factors' influencing the propensity to consume is 'changes in expectations of the relation between the present and the future level of income' (p. 95).

[2]For Keynes' views on the issue see for example Keynes, 1973B, pp. 10–20.

continue, except in so far as there are definite reasons for expecting a change (Keynes, 1973A, p. 51).

Secondly, it is worth restating the obvious that when the information which conditions expectations is wider than the relevant variable's own past history, the rejection, or modification, of adaptive expectations assumptions does not in any way justify the extreme information assumptions of the rational expectations hypothesis.

Further implications

It would seem that the simple adaptive model of equation (2.1) is irrelevant to the real issues in dispute. It is clear that criticisms made of it serve neither to undermine Keynesian analysis nor to justify the rational expectations alternative. By emphasising the supposed association of model (2.1) with Keynesian analysis it appears that advocates of the rational expectations hypothesis wish to convey the view that it is naïve expectational assumptions giving rise to systematic errors of anticipation which largely underlie the policy conclusions that derive from Keynesian macro-models.[1] Such a view is totally mistaken. Indeed it may be argued that it is new classical, and not Keynesian, analysis which relies most on assumptions of systematic expectational errors.

This point becomes most apparent in the context of econometric modelling. With model estimation being mainly confined, through data limitations, to using quarterly or annual data points, the issue arises of how best to model the *actual* dynamics of economic behaviour using such discrete observations. Specifically, if consumers (and others) obtain new observations on, say, prices more or less every day and, as a result, modify their expectations of future prices, and if, as the new classicists assume, expectations are formed in such a manner as to reveal no consistent error bias over time, then it would seem that when using annual or even quarterly data points (and therefore implicitly summing over time) the consistent, most useful option is to equate expectations with actual out-turns.

The structure of Keynesian models is such that doing this, of course, does not eliminate the implied scope that exists for policy intervention. In many areas of the Cambridge Multisectoral Model, for example, expected and 'actual' values of annual data points are implicitly assumed to be equal. (Short run expectations are usually modelled explicitly only where people are assumed to distinguish 'normal' movements in variables in order to avoid instant adjustment for stability reasons—e.g. in industrial price determination (Coutts *et al.*, 1978) or the adjustment of consumer spending.)

For the new classicists the equating of expectations with actual out-turns is more problematic. At some level of aggregation (over time) they need to postulate the existence of incorrect expectations. It is the monetarist proposition that aggregate supply fluctuations are initiated by informational errors (see the so-called Lucas–Sargent proposition in, for example, McCallum, 1977, p. 627). These errors are assumed to hold only for the 'short run', although it would seem that the short-run needs to be greater than the interval between published data points. In other words for new classicists to assume the existence of expectational errors in annual or quarterly or even monthly models, the implicit assumption must be that the daily pattern of expectation errors involves consistent or systematic bias. (The example of econometric modelling has been used because it is the most relevant context and relates to practical problems.

[1]This point of view is doubtless encouraged by those economists who, rather perversely, call themselves Keynesian and yet apparently believe in some form of (expectations augmented) Phillips curve.

The point can be put more straightforwardly however: if for any time period it is assumed that expectations are incorrect (as the new classicists do) then it follows that over the intervening subperiods expectations must be systematically incorrect.)

Thus without assuming systematic errors (at some level of timewise aggregation) monetarists would find business cycles even harder to 'explain' (Lucas, 1975). The Keynesian theory of effective demand has no such difficulties.

3. A question of metholology

It has been argued that certain expectational assumptions that are considered by the new classical economists (amongst others) as being inadequate cannot, as often supposed, legitimately be labelled Keynesian. Keynesians are (or should be) at least as critical of such assumptions. In this section a similar argument is developed in relation to controversies about the estimation of macro-models. An approach to model estimation which we shall refer to as the 'standard' approach but which is frequently (and we shall argue erroneously) labelled Keynesian, forms the object of the Lucas (1976A) critique and justifies, to new classical economists, the rejection of Keynesian econometric models.

In particular Keynesian model building is contrasted with the 'scientifically more demanding' (Lucas, 1976A, p. 42) approach of the advocates of the rational expectations hypothesis. Indeed we are even warned against attempting 'to refine the jerry-built structures to which Keynesian macroeconomics had led us' (Lucas, 1976B, p. 2). Here, however, we shall argue that the approach being criticised, far from being Keynesian, is actually something which is subject to still more fundamental criticism from a Keynesian standpoint.

Alternative econometric approaches

In order to facilitate an assessment of the Lucas-inspired criticisms of Keynesian methodology it is convenient to highlight a distinction between alternative philosophical approaches to model estimation (discussed in detail below). One approach—which for the purpose of this essay will be labelled the 'standard' approach—accepts the existence of 'correctly specified' equations and 'true' models and assumes that the role of the econometrician is to utilise existing data in order to identify such structures.[1] The second approach—which we label Keynesian—essentially rejects the existence of 'true' or 'correct' models and concentrates on obtaining tolerable, data-consistent representations of, or approximations to, *a priori* theory.

The existence of these differences of method seems to have been largely ignored in published writings (at least in explicit form). This may partly explain why even ex-Keynesian converts to monetarism (e.g. Willes, 1980, p. 83) as well as monetarist practitioners (e.g. Minford, 1980, p. 4) seem to have misunderstood, and therefore expected different results from and developments in, Keynesian modelling.

The 'standard' approach

The importance of the distinction between the Keynesian and the standard approach can be usefully illustrated by referring to developments within the latter which have taken place recently, at least within the context of the UK.

[1]As discussed below, notions of true structures tend, in practice, to be assumptions of stable parameters and white noise error processes.

These developments, sometimes referred to as mis-specification analysis[1] (e.g. Hendry, 1979A) and pioneered by Hendry and others (see for example Davidson *et al.*, 1978; Hendry, 1978 and 1979B) amount to assuming that the economy is usually away from, but is constantly tending towards some steady state (or equilibrium) situation. For modelling economic behaviour the procedure proposed is to specify the most general form of equation possible given data limitations (including numerous lag variables and an error-correction term[2]) in order to capture out-of-equilibrium dynamics. Economic theory is assigned the role of determining steady-state restrictions which can be imposed and 'tested', whilst statistical techniques determine everything else there is to know.

This procedure is proposed as a response to the proliferation of equations marked by the relative instability of their parameter estimates and predictive failure. On reading the various contributions by Hendry and others, it is difficult not to be aware of the heavy emphasis on tests of, or ways of increasing, parameter stability and predictive accuracy. This emphasis reflects an underlying belief that there exists a correct equation which needs only to be identified and which apparently possesses constant and stable parameters and usually a white noise error process.[3] The proposed procedure is intended to provide a method of identifying this correct specification. In the terminology of the authors (see Davidson *et al.*, 1978) the procedure is supposed to determine the 'true' structural equations (p. 691) or equivalently which relationships are 'autonomous', and which are 'mere good approximations' (p. 684).

The Keynesian approach

The Keynesian approach to modelling, by contrast, accepts that the *best* that can be obtained are 'mere good approximations'. The presumption is neither that the models are in some formal sense 'true' nor even that there is a possibility of a true model in equation form. Rather theories are abstracted from all types of information about how the world works, and are confronted with data in order to obtain formal representations which are tolerably consistent with the evidence of national accounts and other statistics about the past. The main objective is to construct a tool to aid thinking about current problems and issues.

Prolonged stability of parameter estimates is not an *a priori* expectation and therefore plays no rôle as a criterion to determine correctness of specification, etc. Indeed forecasts are not made on the assumption that identified forcing (supposedly 'independent') variables which have been important in recent history will always be *the* important influences. Forecasts are calculated merely as projections *conditional* upon particular data representations (although certain institutions etc. may treat them otherwise). They are merely an extra source of information to *guide* policy-related decisions and analysis.

It is continuously recognised, moreover, that the real-life interdependence of variables is not fully modelled and that unquantifiable influences will be of importance—sometimes of major importance. Nor are the possibilities of major 'structural changes' discounted. Thus numerous adjustments and qualifications are typically appended to analyses, and the whole process of modelling is conceived of in terms of a continuous evolution, responding to changing spheres of interest and understanding.

[1]Mis-specification is usually defined as 'excluded variables and incorrect exogeneity assumptions' (Hendry, 1978, p. 7).

[2]See for example Hendry, 1978, p. 20.

[3]Doubtless we will at some stage hear about structurally stable 'false' models, etc.

Given the nature of this approach, Keynesian publications tend not to be highly technical or formal and model projections usually represent only part (and not necessarily even a major part) of the analysis (see e.g. *Cambridge Economic Policy Reviews*). The more formalised parts of analyses are usually relegated to appendices, and historical research including case studies will usually be at least as important a part of any analysis as econometric evidence (see for example Wilkinson, 1981).

Some implications

The contrast between the two approaches becomes particularly sharp if we consider the different reactions to breakdowns of previously estimated econometric relationships. Hendry's response to the failure of econometricians to identify stable equations is to emphasise that such 'instability . . . provides little evidence about the stability or otherwise of the sought after econometric relationships . . .' (Hendry, 1978, p. 1). Keynesians with no prior expectation of prolonged stability of parameter estimates typically reationalise the predictive failure of an econometric system on the grounds that some structural break has occurred. Hendry with a prior belief in the existence of a correct model, views this response, in most situations, as 'vacuous and very unconvincing' (Hendry, 1973, p. 3).

Yet the notion of a 'correct' model or equation (and the usual presumption that this will involve stable parameters and a white noise error process) is, of course, no more than an act of faith or an assumption. On adopting this assumption the economist, as we have seen, is reduced to the rôle of attempting to 'test' the most generalised equation, imposing constraints only in so far as they represent notions of long-run, or steady state, behaviour and leaving the rest to statistical techniques. (Compare this with the view that the long run is but the sum of successive, different, short runs.)

If, however, the existence of a 'correct' model is *not* assumed, then economic theory and judgement play a more substantial, if sometimes informal, rôle. There are no *a priori* reasons to expect long-run structural stability (constant and stable parameter 'estimates') in any equations whatsoever. Thus Barker (1976) in describing an early version of the Keynesian Cambridge Growth Project's model notes that '. . . to rely on past data to "speak for itself" is an extremely restrictive approach and rules out consideration of major structural change. Complete reliance on historical time-series would pre-empt policy from ever tackling new problems, or old ones which had not been solved' (p. 6). For similar reasons Cripps and Fetherston (1978), in outlining the methodology adopted by the Cambridge Economic Policy Group (CEPG) warn that 'time series estimation on its own must . . . be considered something of a last resort and should be constrained as tightly as possible by other types of evidence' (p. 10). Clearly Keynesians accept a much more active rôle for economists in modelling the economy and in addressing policy issues generally.

The reason for including the above outline is merely to highlight the fact that there exist at least two methodological traditions. The standard approach which accepts the existence of a 'true' model—that reality has a correct representation in equation form—and the Keynesian approach which, essentially, does not.

The rational expectations critique of existing econometric practice

We are now in a position to assess the relevance of the Lucas (1976A) critique of model building practice.

In the first stage of his thesis Lucas highlights examples of 'current' econometric practice which according to him are at variance with econometric theory. He singles out in particular the 'indifference of econometric forecasters to data series prior to 1947' (p. 22), the 'frequent and frequently important refitting of econometric relationships' (p. 23) and 'the practice of using patterns in recent residuals to revise intercept estimates for forecasting purposes' (p. 23).

These examples are part of the evidence cited by Lucas to facilitate the inference that although models so estimated may be successful at short-term forecasting the continual process of model adjustment involved implies that they may not be useful for forecasting longer-term outcomes. Indeed Lucas suggests that 'the "long-run" implications of current forecasting models are without content' (p. 24) and 'that simulations using these models can, in principle, provide *no* useful information as to the actual consequences of alternative economic policies' (p. 20).[1]

The second stage of Lucas' work attempts to identify why examples should exist which are at variance with 'the standard, stable-parameter view of econometric theory and qualitative policy evaluation' (p. 24). His analysis leads him to conclude that for any equation, existing parameter estimates are only relevant so long as the particular policy that pertained during the sample period, continues to do so. As soon as a policy changes so too does the process of individual decision making so that 'any change in policy will systematically alter the structure of econometric models' (p. 4).

The rational expectations solution

It is clear, therefore, that Lucas' work is motivated by similar apparent failings of (the standard) econometric practice to those noted by Hendry. The proposed solutions, however, diverge widely. In contrast to Hendry, Lucas explicitly doubts the usefulness of 'disequilibrium dynamics' (p. 19) and 'hesitates to indulge the common illusion that "general" structures are more useful than specific, empirically verified ones . . .' (p. 40). However he shares the common presumption that there exists a 'correct' or 'true' model needing only to be identified. Indeed Lucas explicitly points out that his contentions are:

> based not on deviations between estimated and 'true' structures prior to a policy change, but on deviations between the prior 'true' structure and the 'true' structure prevailing afterwards (p. 20).

The solution which Lucas begins to formulate (although seemingly with less enthusiasm than many contributors since (see for examples Willes, 1980)) is, of course, the rational expectations hypothesis, suggesting, at least in certain cases, that it is 'while scientifically more demanding, entirely operational' (p. 42). Similarly Sargent and Wallace (1970) justify their adoption of the rational expectations hypothesis by arguing that it 'is consistent with the findings that large parts of macro-econometric models typically fail tests for structural change (essentially versions of the Chow Tests)' (p. 179). The implication at least is that *if* structural breaks in existing models occur because policy changes lead to (unmodelled) revisions of decision processes, then the explicit modelling

[1]As an aside it can be noted, however, that although this part of Lucas' argument may seem compelling it is logically not conclusive. If, for example, an equation is estimated using data covering the last million years and it turns out to reveal remarkable parameter stability when estimated over any subsample, this finding does not allow us to conclude that the estimated equation will forecast tomorrow's outcome sufficiently accurately (although we might expect it to). Similarly the existence of a process of equation revision as discussed by Lucas does not mean *per se* that current estimates will not maintain their relevance for the distant future.

of decision making, with expectations formed with knowledge of, and conditional upon, relevant policy régime(s), can provide parameter estimates which are stable, and indeed a model which is 'correctly specified'.

The relevance for Keynesian analysis

Although the Lucas critique is typically addressed to Keynesian analysis, it should by now be clear that the criticism is, in the main, misdirected. The first stage of the Lucas critique which notes the divergence between (standard) econometric theory and econometric practice, based as it is upon considerations of stable estimates and so on, can only be relevant for those who have already accepted the assumption of the existence of a 'correct' or 'true' set of equations. Lucas' and Hendry's arguments may well represent valid criticisms of each other's contributions because they are both made of, and within, the same basic framework or approach. But the Keynesian methodological approach is, as we have seen, fundamentally different, and cannot be criticised in the same terms.

The second part of Lucas' critique is that a change in government policy may lead to structural change in the economy necessitating the re-estimation of econometric relationships. Lucas appears more prepared to accept this conclusion than Hendry although he believes that an alternative solution may be attempted through assuming that people can anticipate government policy régimes and therefore adjust to structural changes. The implication is that the assumption of rational expectations may lead to stable parameter models.

The important point is that once again the critique does not represent a valid criticism of Keynesian modelling practice. Indeed the latter is more consistent with an extension of Lucas' criticisms. Keynesians may suspect that following a structural change of any kind, and not just a change in government policy régimes, many aspects of their models, both including but also in addition to expectations assumptions, will need to be revised and reformulated.[1] The latter would only definitely *not* be the case if apart from expectations assumptions, the models were 'correctly' specified. And this is exactly what economists in the new classical school, not Keynesians, assume.

Thus, as with expectations theories discussed in Section 2, new classical arguments concerning econometric methodology appear to be misleading. Not only are they misdirected if seen as criticisms of Keynesian modelling practice, but they can only be relevant if addressed to econometric practice which adopts the same general approach as that of the new classicists. Existing practice can in fact be equally, and perhaps more substantially, criticised from a Keynesian standpoint.

Some 'true model' implications

Having raised the above methodological issues it is worth-while drawing out some implications of the true model approach. The true model assumption plays a part in allowing monetarists to derive policy conclusions concerning the neutrality, or ineffectiveness, of government policy. In this framework steady state restrictions can be imposed in accordance with the 'natural rate hypothesis' and other monetarists assumptions. These specify 'equilibrium positions' from which departures occur only on account of informational errors. The rational expectation assumption that people

[1]See for example Kilpatrick and Lawson (1980) for some qualifications that may be required if import controls of some form are imposed.

are aware of the 'true' model to within a white noise error term underwrites the neutrality propositions by ensuring the absence of systematic 'disequilibrium'.

A second implication of the 'true model' approach is that if its proponents are to construct forecasting models—and they do—then if the expectations mechanisms within them are not to be a continuous and systematic source of inaccuracy, and given that rational expectations are formed conditionally upon the structure of the entire model, it follows that the formalised model has to be the 'correct' one. Thus the process of predictive failure should help monetarists determine which models to reject. This point seems to be behind Minford's (1980) prediction of a future convergence in model structures so 'that within five years the standard macroeconomic model will be an equilibrium model in which expectations are formed optimally . . .' (p. 24). An alternative hypothesis, of course, is that all rational expectations models will be rejected.[1,2]

Summary and qualifications

Two different approaches to econometric modelling have been identified. The first assumes that there exists a 'true' or 'correctly' specified set of equations which have only to be identified. The contributions of both Hendry *et al.* and rational expectations proponents including Lucas are within this framework. Given that each makes criticisms of existing econometric practice within this framework or approach the implications are less relevant for work which adopts the Keynesian approach.

The Keynesian approach to modelling does not accept that reality has a 'correct' representation in equation form. A model is seen as a 'mere approximation' to economic behaviour and is viewed mainly as a tool to aid thinking and so further analysis. Keynesian models are 'intended to evolve in the light of experience and criticism and in response to the changing focus of interest in specific policy objectives and instruments' (Cripps and Godley, 1976, p. 335). Whilst pretentions to scientific precision are not attempted nor even puzzled over (Hendry, 1980), and concepts of 'true' models are not invoked, the emphasis of any analysis is on being as realistic as possible. Thus, for example, the 'ultimate criterion adopted by the CEPG is that its analysis should be judged in terms of success or failure in giving realistic assessments of the main contemporary policy issues which arise for the UK economy as a whole' (Cripps and Fetherston, 1978, p. 2). In this process, however, equations are viewed as no more than serviceable approximations to current behaviour, and just one of many forms of information utilised to this effect.

Doubtless there will be economists who consider themselves Keynesian but who object to some of the views presented above. Certainly these views reflect the author's own

[1]See Hahn (1980) for an interesting discussion of reasons why we should not expect equilibrium models to be so pervasive.

[2]There are, of course, various other implications of the 'true model' approach which are less relevant for the purposes of this paper. An interesting one concerns the normative consequences, which do seem to have been observed by Sargent and Wallace (main protagonists of the rational expectations hypothesis) but left as an unsolved 'dilemma'. They write:

> The conundrum facing the economist can be put as follows. In order for a model to have normative implications, it must contain some parameters whose values can be chosen by the policy maker. But if these can be chosen, rational agents will not view them as fixed and will make schemes for predicting their values. If the economist models the economy taking these schemes into account, then those parameters become endogenous variables and no longer appear in the reduced form equations for the other endogenous variables. If he models the economy without taking the schemes into account, he is not imposing rationality (Sargent and Wallace, 1976, p. 183).

position. I feel justified, however, in referring to the approach as Keynesian if only because it seems so much in the spirit of Keynes, as the following passage perhaps illustrates:[1]

> The object of our analysis is not to provide a machine, or method of blind manipulation, which will furnish an infallible answer, but to provide ourselves with an organised and orderly method of thinking out particular problems; and after we have reached a provisional conclusion by isolating the complicating factors one by one, we then have to go back on ourselves and allow, as well as we can, for the probable interactions of the factors amongst themselves. This is the nature of economic thinking. Any other way of applying our formal principles of thought (without which, however, we shall be lost in the wood) will lead us into error (Keynes, 1973A, p. 297).

Although the Keynesian approach is clearly preferred by the author it has not been the intention here to justify its adoption (see for example Cripps and Fetherston (1978) for a more substantial contribution of this sort). The thesis of this section is that Keynesian methodology should be viewed as an alternative, not a predecessor, to that of the proponents of the rational expectations hypothesis. The difference represents not a 'technological lag' but a conscious and informed choice. It is an alternative methodology that is not susceptible to the critique of econometric practice advanced by Lucas and others. Indeed we have argued that the econometric modelling practice which Lucas focuses upon can be more substantially criticised from a Keynesian standpoint.

4. Conclusion

This paper has developed the following single theme: that numerous arguments put forward to undermine Keynesian analysis and model building and, by implication at least, to 'justify' the adoption of rational expectations models, have been fundamentally misleading and mistaken.

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[1]See also Keynes' responses (and related correspondence) to Tinbergen's early work (Keynes, 1973D, pp. 285–320).

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Appendix

(1) First consider the case (generalised below) where Y_t does not exhibit a trend:

$$Y_t = Y_{t-1} \tag{A1.1}$$

A straightforward and general way to introduce uncertainty is to reformulate this 'system' as follows:

$$\begin{aligned} Y_t &= \pi_t + \varepsilon_t \\ \pi_t &= \pi_{t-1} + \gamma_t \end{aligned} \tag{A1.2}$$

where π_t is the deterministic part of Y_t and ε_t and γ_t are both independently distributed stochastic processes. For simplicity, and lacking contrary evidence, we assume, as usual, that all shocks and disturbances are distributed normally, so that

$$\begin{aligned} \varepsilon_t &\sim N(0;\ V_{\varepsilon t}) \\ \gamma_t &\sim N(0;\ V_{\gamma t}) \end{aligned} \tag{A1.3}$$

Writing the density function of π_t at $t-1$ as $N({}_{t-1}\pi_t^e;\ \Sigma_t)$ so that ${}_{t-1}\pi_t^e$ (in the notation of Section 2) is the prior (mean estimate) of Y_t it can be demonstrated using Bayes Theorem (for proof see Lawson, 1980, p. 316) that the prior (mean) estimate of Y_{t+1} is given by:

$${}_t\pi_{t+1}^e = {}_{t-1}\pi_t^e + \lambda_t(Y_t - {}_{t-1}\pi_t^e) \tag{A1.4}$$

where

$$\lambda_t = \frac{\Sigma_t}{\Sigma_t + V_{\varepsilon t}} \tag{A1.5}$$

and so the adjustment coefficient is time varying. The updated prediction variance term is

$$\Sigma_{t+1} = V_{\gamma t} + \Sigma_t - \frac{(\Sigma_t)^2}{V_{\varepsilon t} + \Sigma_t}$$

Thus an unconstrained or 'generalised' adaptive expectations mechanism has been derived.

(2) In the more general case where Y_t reveals a trend, systems (A1.2) and (A1.3) can be reformulated as follows:

$$\begin{aligned} Y_t &= \pi_t + \varepsilon_t & \varepsilon_t &\sim N(0;\ V_{\varepsilon t}) \\ \pi_t &= \pi_{t-1} + g_t + \gamma_t & \gamma_t &\sim N(0;\ V_{\gamma t}) \\ g_t &= g_{t-1} + \delta_t & \delta_t &\sim N(0;\ V_{\delta t}) \end{aligned} \tag{A2.1}$$

Transforming to matrix notation we define

$$\mathbf{P}_t = \begin{pmatrix} y_t \\ g_t \end{pmatrix} \qquad \boldsymbol{\alpha}_t = \begin{pmatrix} \gamma_t \\ \delta_t \end{pmatrix}$$

and

$$\boldsymbol{\alpha}_t \sim \mathrm{N}\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix};\ \mathbf{F}_t\right]$$

so that

$$\mathbf{F}_t = \begin{pmatrix} V_{\gamma t} & 0 \\ 0 & V_{\delta t} \end{pmatrix}$$

Let

$$\mathbf{S}_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \qquad \mathbf{S}_2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \qquad \mathbf{X} = (1 \quad 0)$$

so that system (A2.1) may be rewritten:

$$\begin{aligned} Y_t &= \mathbf{X}\,\mathbf{P}_t + \varepsilon_t & \varepsilon_t &\sim N(0;\ V_{\varepsilon t}) \\ \mathbf{P}_t &= \mathbf{S}_1\,\mathbf{P}_{t-1} + \mathbf{S}_2\,\boldsymbol{\alpha}_t & \boldsymbol{\alpha}_t &\sim N(0;\ \mathbf{F}_t) \end{aligned} \tag{A2.2}$$

If the mean expectations of $\mathbf{P}_t$ held at $t-1$ is ${}_{t-1}\mathbf{P}^e{}_t$, with variance ${}_{t-1}\boldsymbol{\Sigma}_t$, then the posterior, or updated, estimate of $\mathbf{P}_t$, following the observation of Y_t, is ${}_t\mathbf{P}^e{}_t$ where:[1]

$${}_t\mathbf{P}^e{}_t = {}_{t-1}\mathbf{P}^e{}_t + \boldsymbol{\lambda}_t(Y_t - \mathbf{X}\ {}_{t-1}\mathbf{P}^e{}_t) \tag{A2.3}$$

[1]For proof see Graybill (1961), Theorem 3.10, p. 63.

The adjustment coefficient is given by:

$$\boldsymbol{\lambda}_t = {}_{t-1}\boldsymbol{\Sigma}_t\, \mathbf{X}'\, (\mathbf{X}\, {}_{t-1}\boldsymbol{\Sigma}_t\, \mathbf{X}' + W_t)^{-1} \tag{A2.4}$$

and variance of the estimate is

$${}_t\boldsymbol{\Sigma}_t = {}_{t-1}\boldsymbol{\Sigma}_t\, \mathbf{X}'\, (\mathbf{X}\, {}_{t-1}\boldsymbol{\Sigma}_t\, \mathbf{X}' + W_t)^{-1}\, \mathbf{X}\, {}_{t-1}\boldsymbol{\Sigma}_t \tag{A2.5}$$

It follows from (A2.2) that

$${}_t\mathbf{P}^e_{t+1} = \mathbf{S}_1\, {}_t\mathbf{P}^e_t \tag{A2.6}$$

and that the variance of this forecast is given by:

$${}_t\boldsymbol{\Sigma}_{t+1} = \mathbf{S}_1\, {}_t\boldsymbol{\Sigma}_t\, \mathbf{S}_1' + \mathbf{S}_2\, \mathbf{F}_t\, \mathbf{S}_2' \tag{A2.7}$$

Thus an adaptive expectations mechanism has been derived which is immune from the usual criticisms made by the proponents of the rational expectations hypothesis. To make this mechanism operational initial values of $\mathbf{P}_0$ and $\boldsymbol{\Sigma}_0$ must be determined. A procedure for doing this is set out in Lawson (1980).